

Design Assumptions and Planning Phase

1. Assumptions

AHB-Lite System Assumptions

Clocking: All AHB-Lite components operate on a single synchronous clock (`HCLK`). Reset (`HRESETn`) is active-low and asynchronous.

Data Width: The bus is 32-bits wide (`HSIZE` up to `WORD`). Transfers larger than 32-bits are not supported.

Protection & Lock: The `HPROT` and `HMASTLOCK` signals are implemented in the interface but are tied to default values (unsupported/ignored).

Wait States: Slaves can insert an arbitrary number of wait states by driving `HREADYOUT` low. The master must hold `HTRANS` , `HADDR` , `HSIZE` , `HBURST` , `HWRITE` , and `HWDATA` stable during wait states.

Error Responses: A slave error (`HRESP=1`) always spans exactly two cycles as per the AMBA 3 specification. The master will immediately abort the remainder of any active burst upon receiving an error.

SPI Slave Assumptions

Mode: SPI operates in Mode 0 (`CPOL = 0` , `CPHA = 0`). Data is sampled on the rising edge of `SCK` and shifted out on the falling edge.

Transaction Length: Every transaction is exactly 16 bits (8-bit command/address + 8-bit data).

2. Planning Phase

This section details the architectural planning for the AMBA AHB-Lite System and SPI Slave, built strictly following the Analog Devices (ADI) Training Session guidelines.

Architecture Overview

Unlike monolithic designs, this project utilizes a modular, hierarchical approach separating the control logic, datapath, and interconnect into dedicated system components.

Component	File	Responsibility
FSM Controller	<code>ahb_master_fsm.sv</code>	Manages the 5-state Moore/Mealy hybrid machine controlling bus phases.
Address Gen	<code>ahb_addr_gen.sv</code>	Calculates <code>current_addr + increment</code> and handles complex bitwise logic for <code>WRAP4/8/16</code> boundaries.
Burst Counter	<code>ahb_burst_counter.sv</code>	Tracks remaining beats. If <code>SINGLE</code> , finishes immediately. If <code>INCR</code> , runs indefinitely. If fixed burst, stops at required count.
Decoder	<code>ahb_decoder.sv</code>	Combinatorially maps <code>HADDR[11:10]</code> to <code>HSEL</code> for up to 3 slaves.
Multiplexor	<code>ahb_mux.sv</code>	Routes <code>HRDATA</code> and <code>HREADYOUT</code> back to the master based on the latched <code>HSEL</code> .
SPI Slave	<code>spi_slave.sv</code>	16-bit shift register mechanics synchronizing to external <code>sck</code> edges.
Reg Map	<code>spi_reg_map.sv</code>	Maps 4 required internal registers, linked to the 7-bit SPI address field.

1. `ahb_master_fsm.sv`

A dedicated synchronous controller responsible for exclusively managing the AHB-Lite phases:

- Drives `HTRANS` generation for the address phase.
- Monitors `HREADY` for the data phase of the previous beat to track the exact pipelining overlap.

2. ahb_addr_gen.sv

An independent datapath module responsible for address calculation:

- Computes subsequent addresses for `INCR` bursts based on `HSIZE`.
- Applies bitwise masking for strict `WRAP` boundaries to guarantee the address loops securely within the target window.

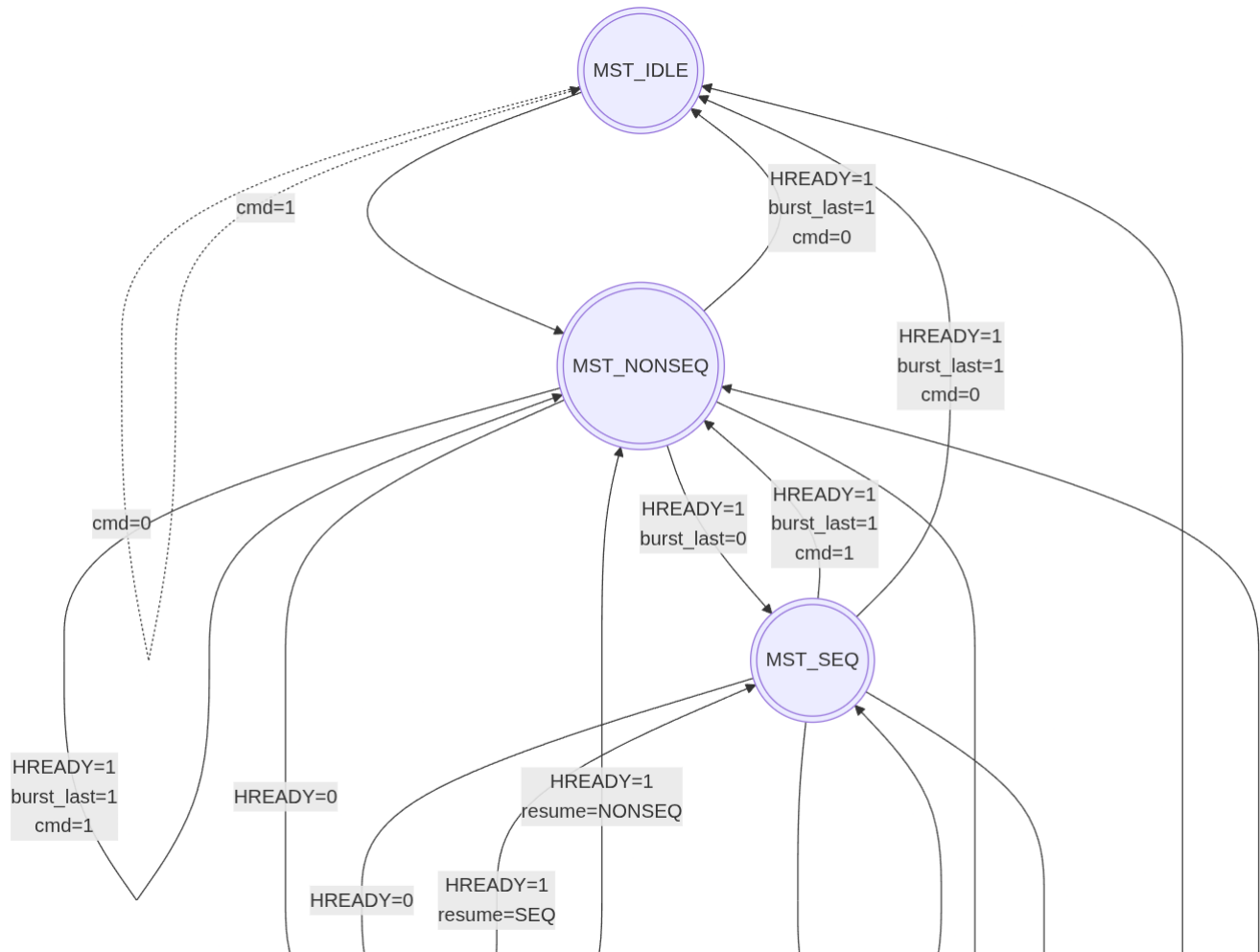
3. ahb_interconnect.sv

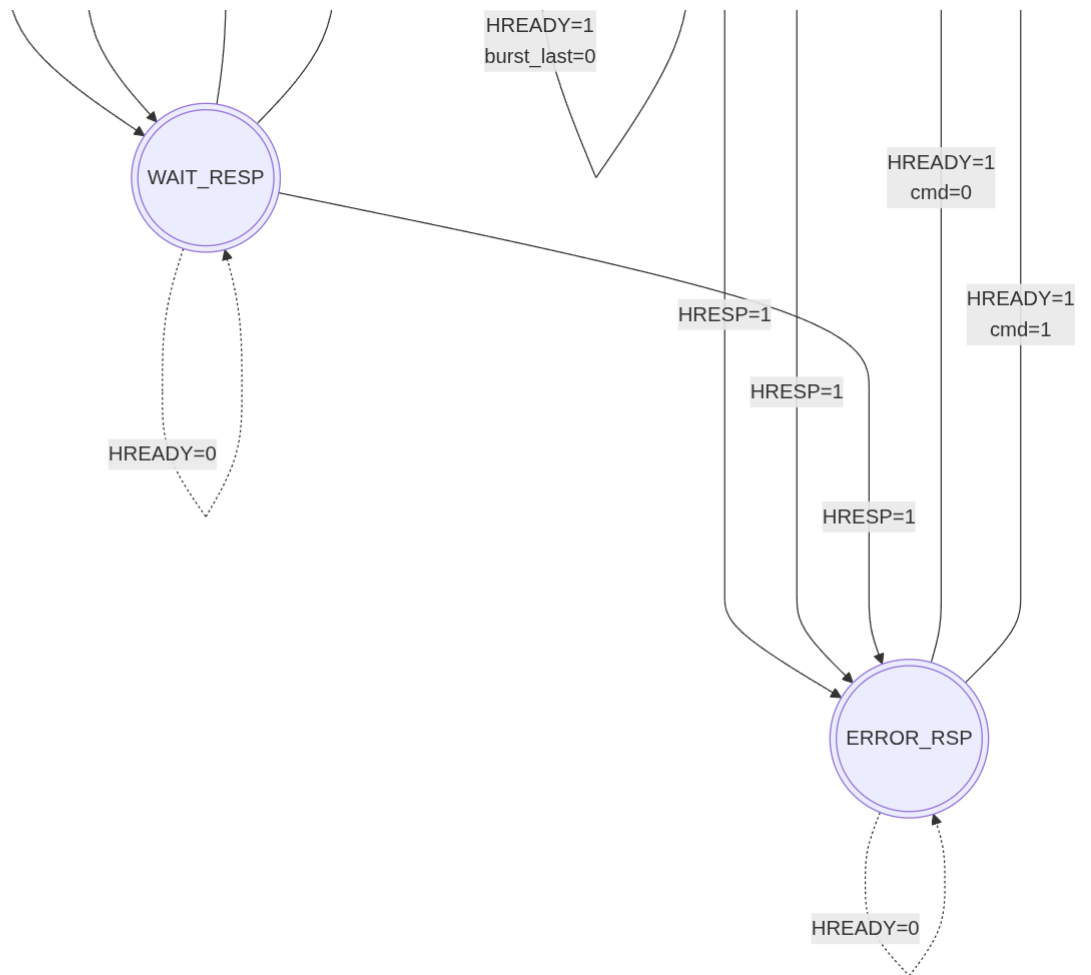
The main routing hub that orchestrates the system by linking the decoder and mux simultaneously:

- Eliminates the need for a single monolithic slave.
- Latches the address phase `HSEL` (gated by `HREADY`) so the data phase routing remains perfectly stable during pipelined overlap.

3. AHB-Lite Master State Machine Model

The following state machine was designed and modeled *before* writing the RTL code. It defines how the master transitions through the AHB phases.





State Descriptions:

MST_IDLE : Default state. Bus is quiet (**HTRANS** = **IDLE**). Waits for a valid user command.

MST_NONSEQ : Initiates the first beat of any transfer. Drives **HTRANS** = **NONSEQ** and the start address.

MST_SEQ : Used for the 2nd to Nth beats of a burst. Drives **HTRANS** = **SEQ** and the sequential (or wrapped) addresses.

MST_WAIT_RESP : Entered if a slave asserts wait states (**HREADY** = 0). The master holds all control signals stable and waits for the slave to finish the data phase.

MST_ERROR_RSP : Entered if the slave asserts **HRESP** = 1 . This is the second cycle of the 2-cycle error response. After this cycle, the master cancels the burst and returns to **IDLE** .